

Tree Borrows

Neven Villani, Johannes Hostert, Derek Dreyer, Ralf Jung

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ETH zürich



MAX PLANCK INSTITUTE
FOR SOFTWARE SYSTEMS



Rust's type system enables powerful optimizations

```
fn write_both(x: &mut i32, y: &mut i32) -> i32 {  
  
    *x = 13;  
  
    *y = 20;  
  
    *x  
  
}
```

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mutable thus disjoint

*x is unchanged

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*x has known value

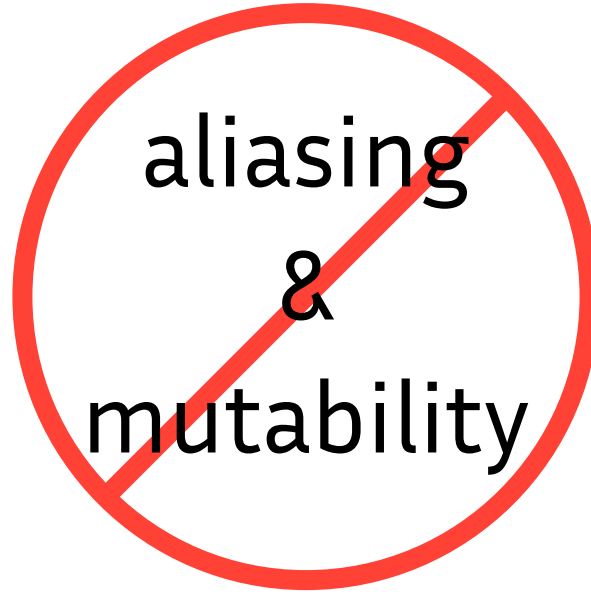
*x is unchanged

always returns 13

Rust's type system enables powerful optimizations

```
fn write_both(x: &mut i32, y: &mut i32) -> i32 {  
  
    *x = 13;  
  
    *y = 20;  
  
    13 // formerly *x: one fewer load from memory  
  
}
```

Type-level guarantees for references



`&mut` → mutation, no aliasing

`&` → aliasing, no mutation

Escape hatch: **unsafe**

Can use **unchecked operations** to do **low-level manipulations**

```
unsafe {  
    // Code within this block can effectively  
    // bypass some parts of the typechecker.  
    ...  
}
```

Within **unsafe** it is **the programmer's responsibility** to check

- that pointers are non-null
- that memory is initialized
- absence of data races
- ...

} violations trigger UB
(Undefined Behavior)

What if **unsafe** code is misused ?

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fn write_both(x: &mut i32, y: &mut i32) -> i32 {
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fn main() {
    let mut root = 42;
    let ptr = &raw mut root;
    let x = unsafe { &mut *ptr };
    let y = unsafe { &mut *ptr };
    println!("{}", write_both(x, y)); // prints 20
}
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```

unsafe code can break the assumptions
that optimizations need!

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Expanding our notion of UB

Within **unsafe** it is **the programmer's responsibility** to check

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- absence of data races
- compliance with aliasing rules ^{NEW!}

} violations trigger UB

Expanding our notion of UB

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Tree Borrows (TB): defines those aliasing rules

Compiler **assumes absence of UB**, exploits this for optimizations

weak optimizations

hard to write correct code



Stacked Borrows (SB)

[Jung et al., POPL'20]

In safe Rust, the Borrow Checker makes borrows well-bracketed. Stacked Borrows extends the well-bracketedness to **unsafe**.

```
let mut root = 42;  
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root

- new stack at root

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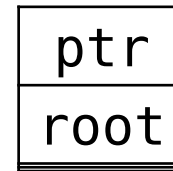
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- ✓ root is at the top

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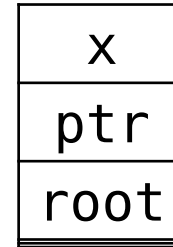


- ✓ root is at the top
- push ptr

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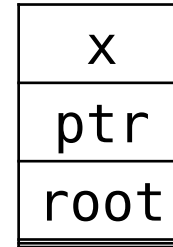


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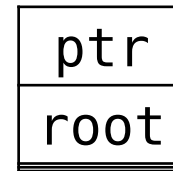


- pop until ptr is at the top

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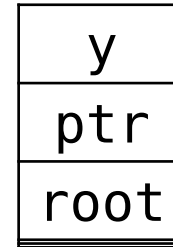


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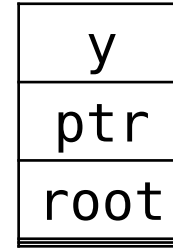


- pop until ptr is at the top
- push y

Desired outcome: UB

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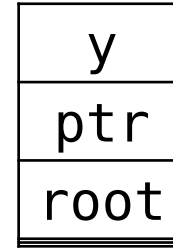
- search for x

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```

UB!



Can't use x if it is not in the stack

Desired outcome: UB

SB was **implemented** in Miri (official interpreter and UB detector)

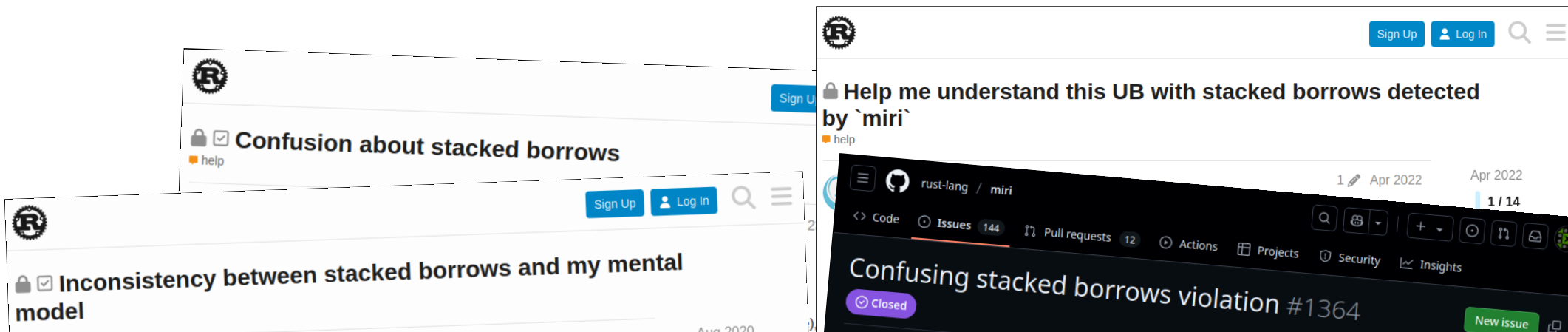
→ included in many projects' CI

→ many bugs detected (e.g. in stdlib)

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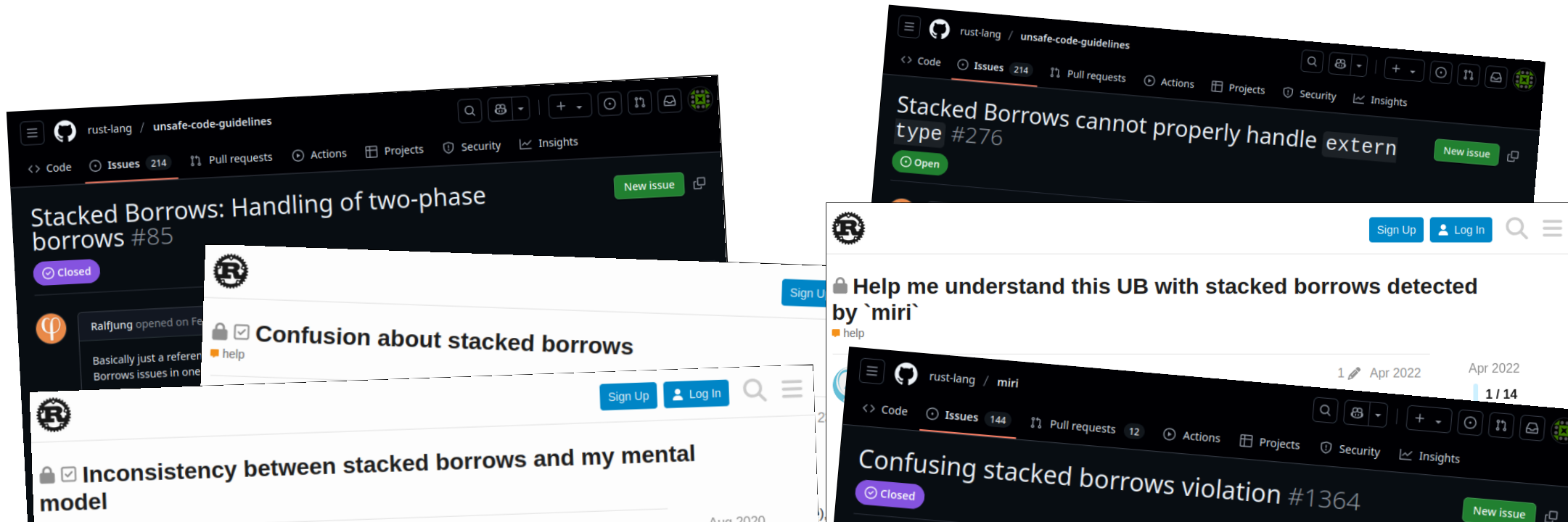
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CD

New issue

Now issue

New issue

liasing-model

New issue

✔ Closed



 help

ⓧ Closed

New issue

Anecdotal evidence supported by data:

- analysis of 30 000 libraries
- 6000+ tests have aliasing UB under Stacked Borrows (leading cause of UB)

Tree Borrows uses a **tree** instead of a stack to track borrows

Out of 30 000 most downloaded libraries,
54% fewer tests with aliasing UB when using Tree Borrows

Tree Borrows uses a **tree** instead of a stack to track borrows

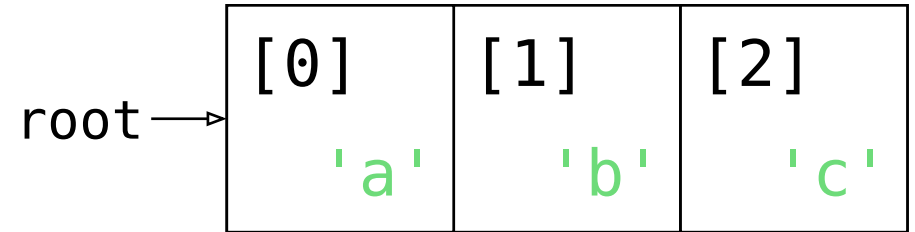
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Fixes known technical limitations of SB,
incl. 2-phase borrows, extern types, **pointer offsets**

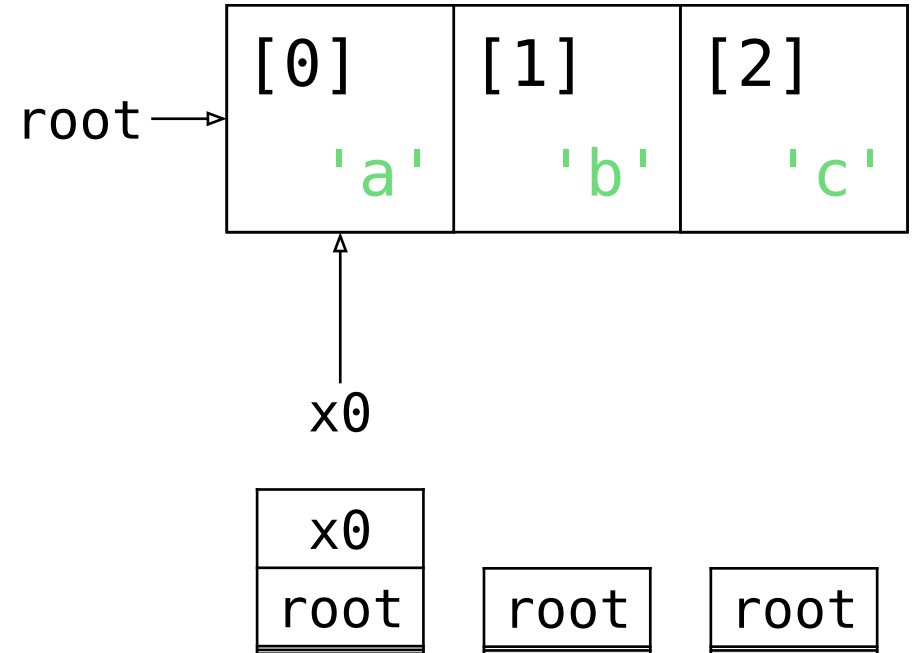
From Stacks to Trees

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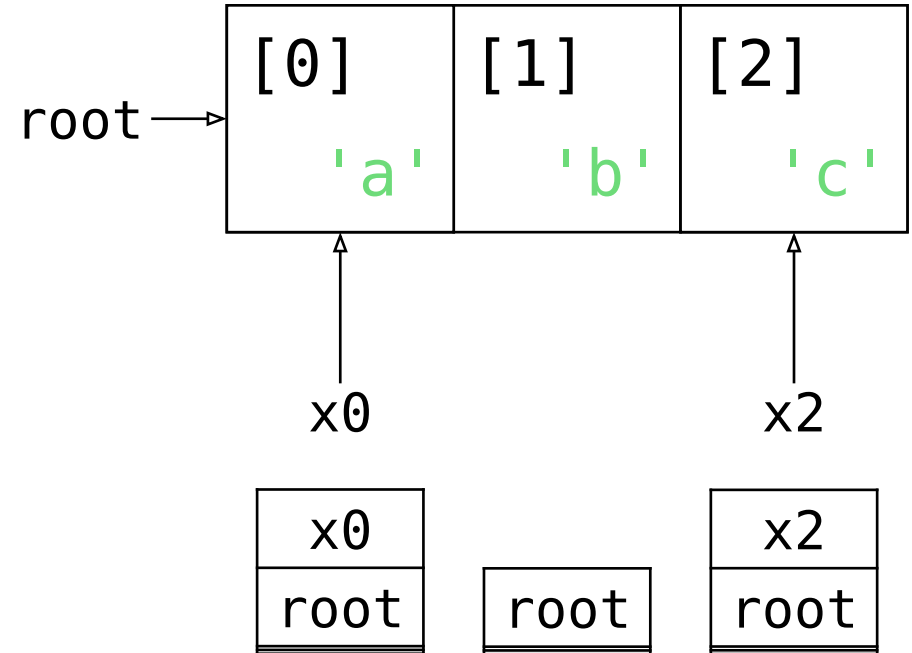
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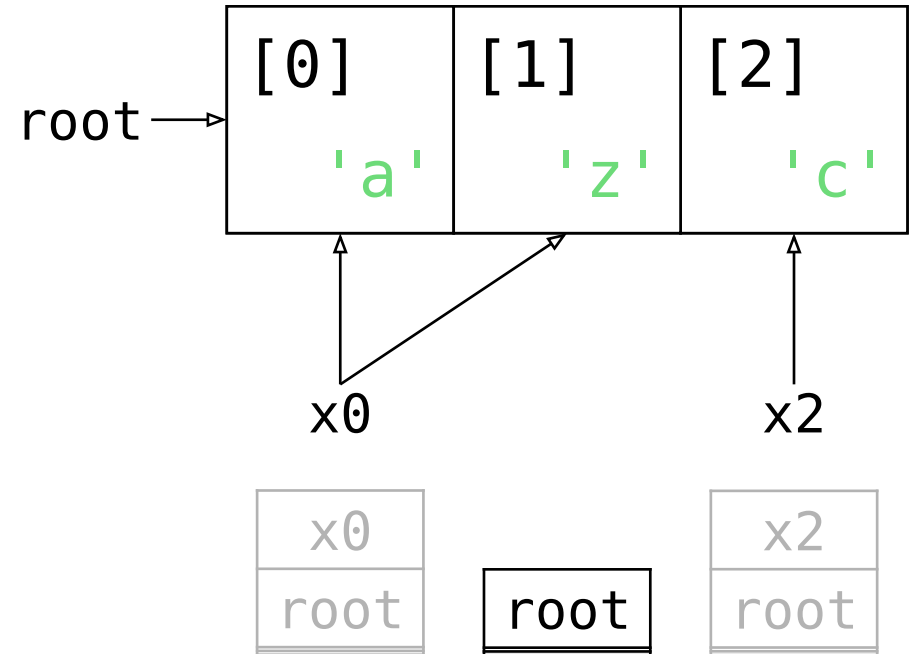
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```
let mut root = vec!['a', 'b', 'c'];
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// Scenario 1
unsafe { *x0.add(1) = 'z'; }
```

Desired outcome: not UB

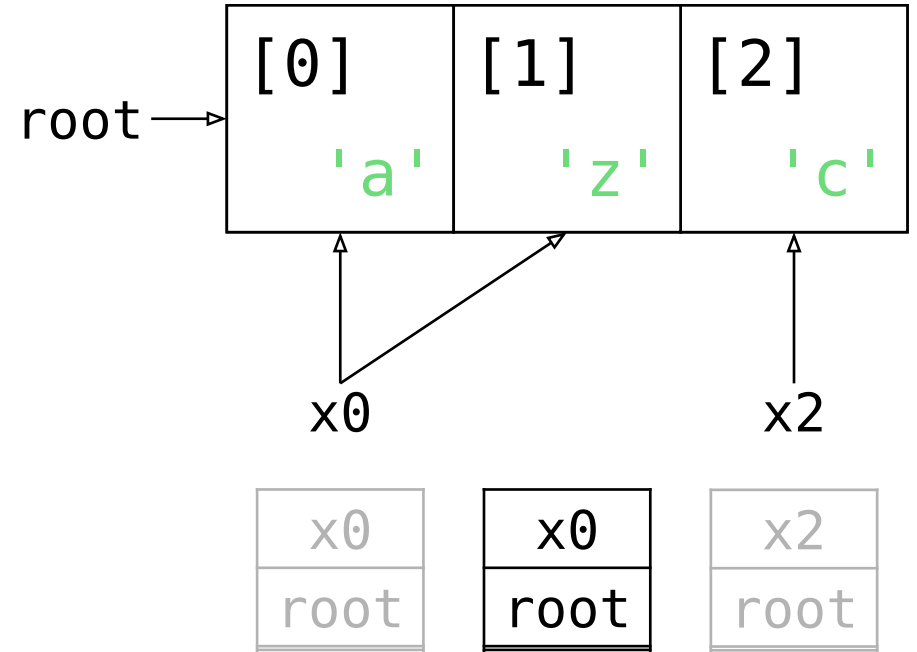


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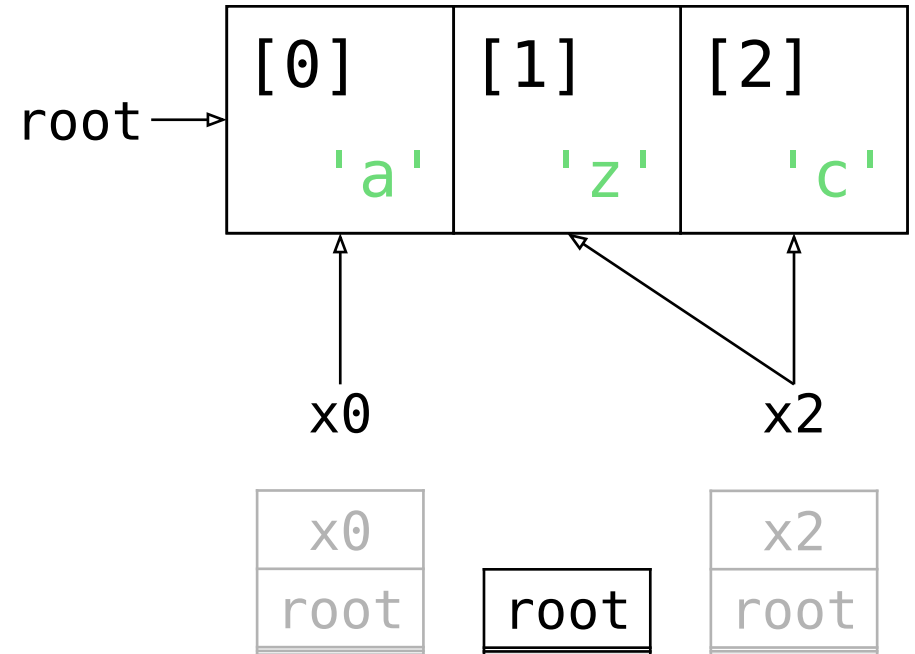


```
let mut root = vec!['a', 'b', 'c'];  
let x0 = &raw mut root[0];  
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```

// Scenario 2

```
unsafe { *x2.sub(1) = 'z'; }
```

Desired outcome: not UB

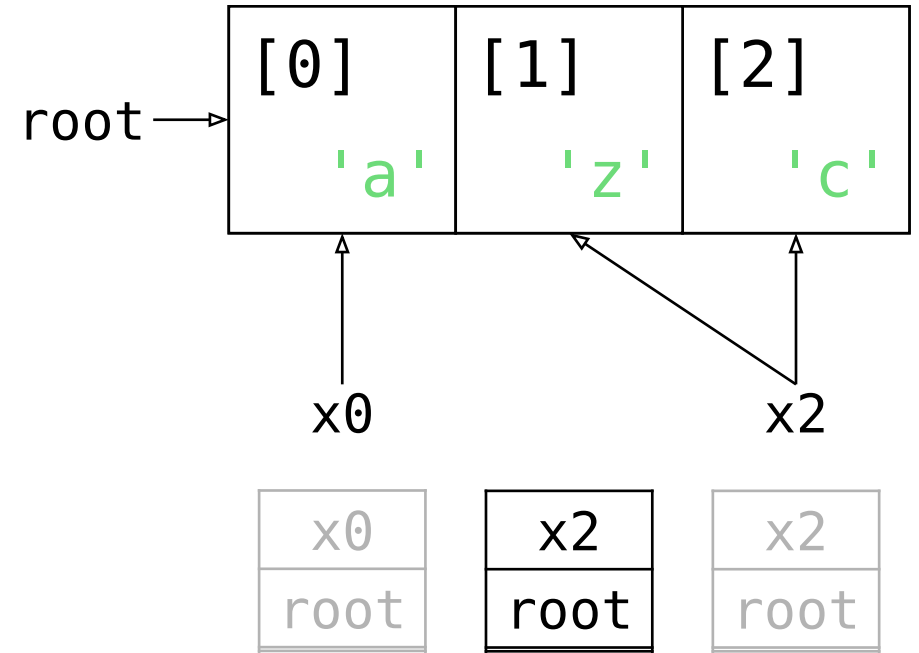


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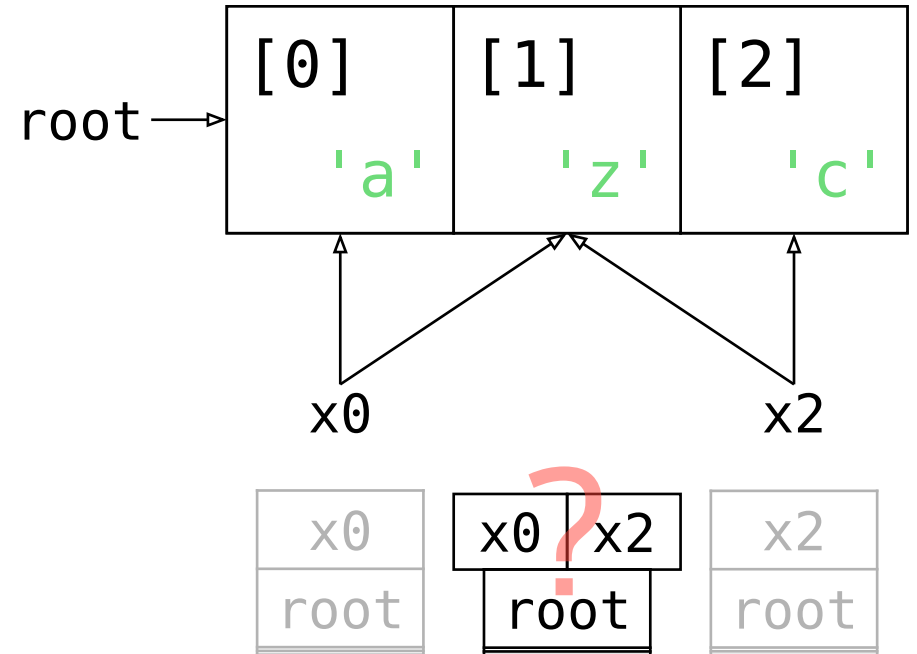
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let mut root = vec!['a', 'b', 'c'];
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// Scenario 1 or 2
unsafe { *??? = 'z'; }
```

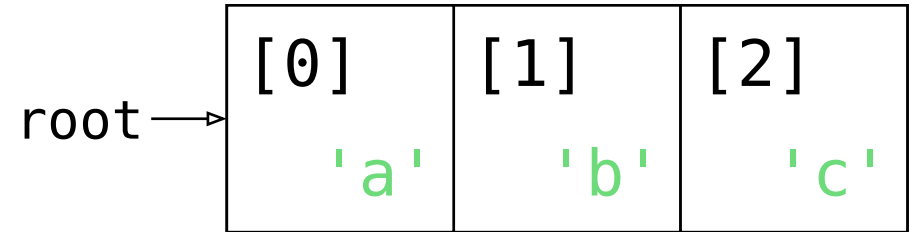
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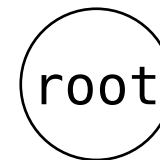
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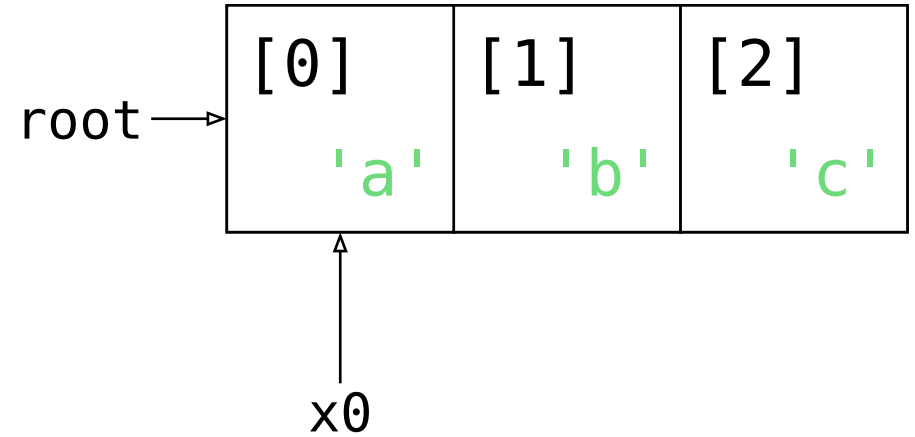
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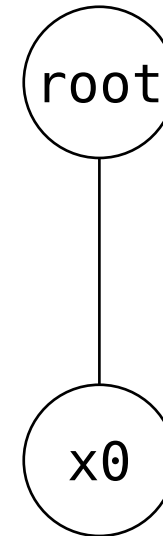
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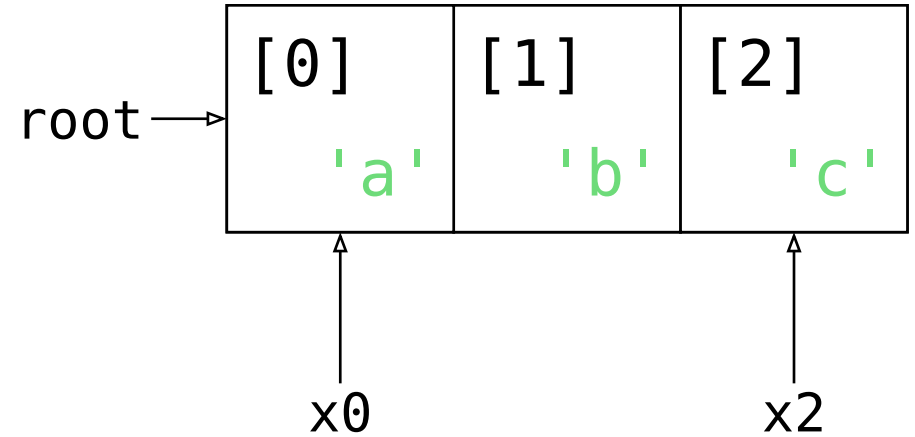
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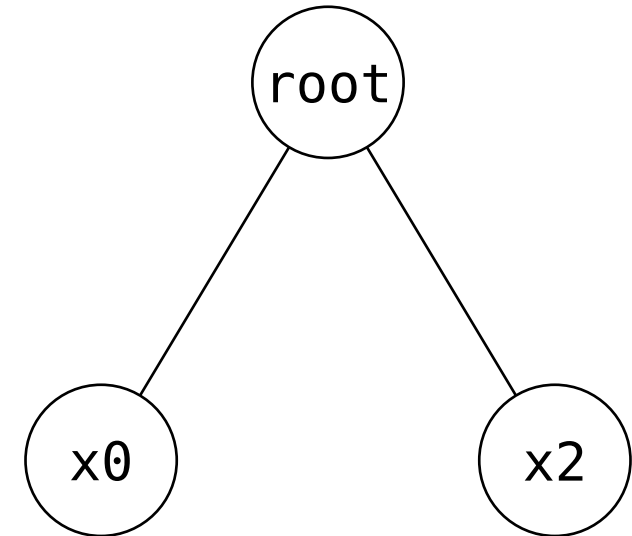
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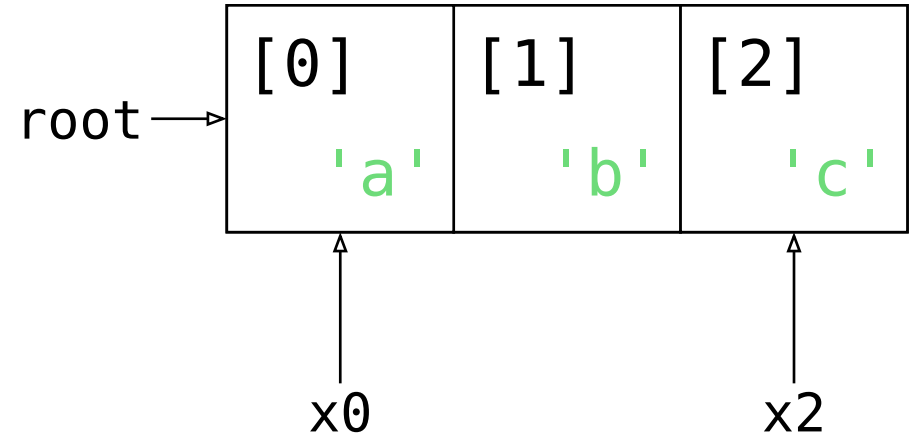
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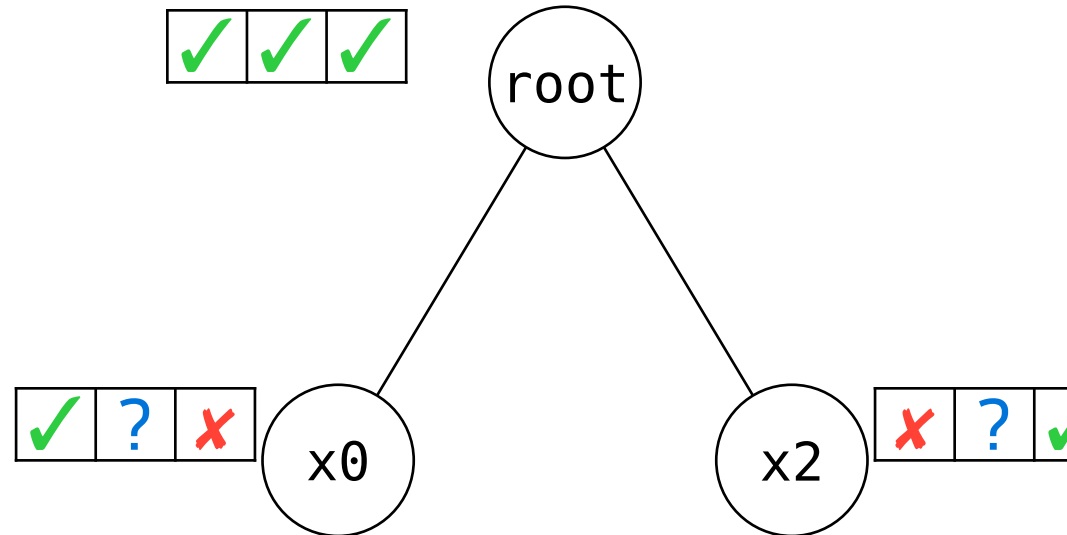
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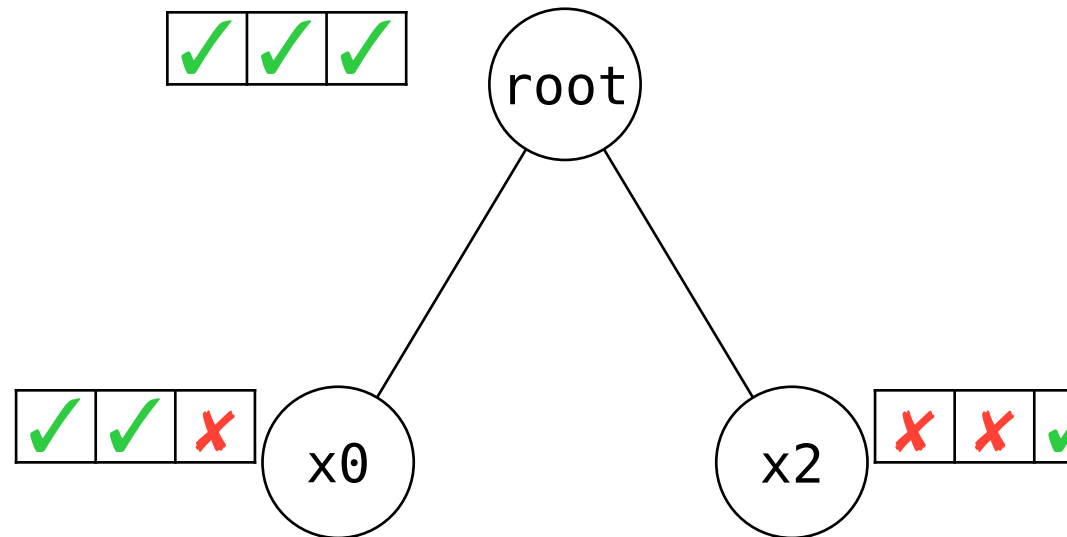
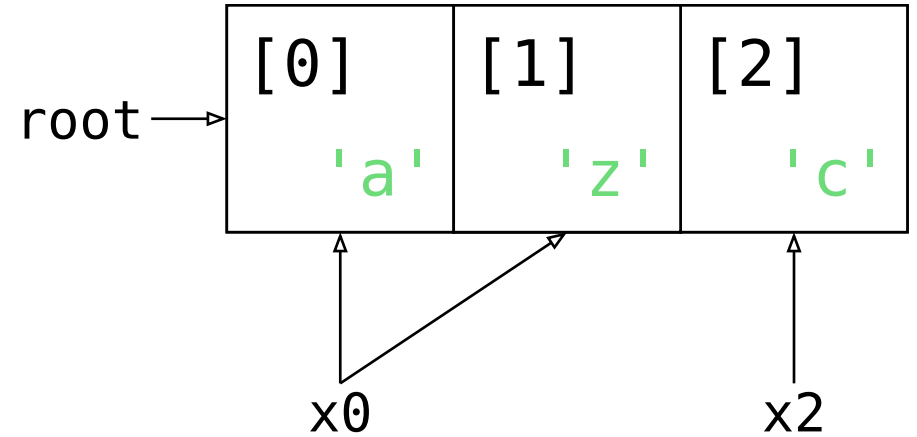


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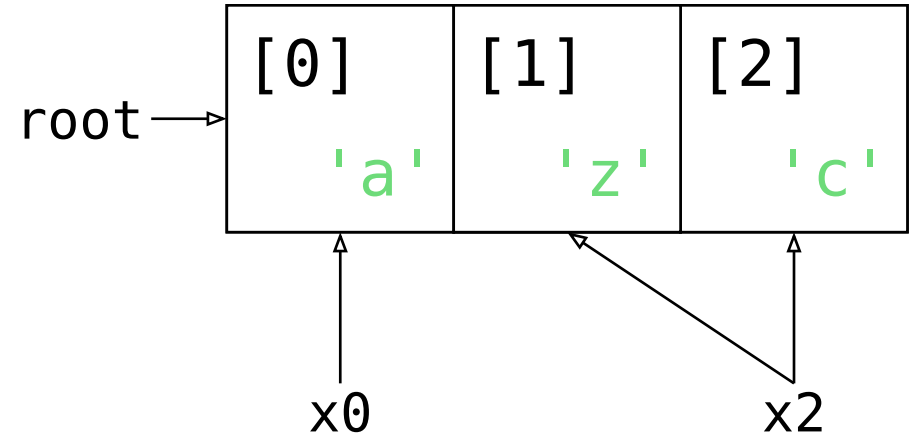
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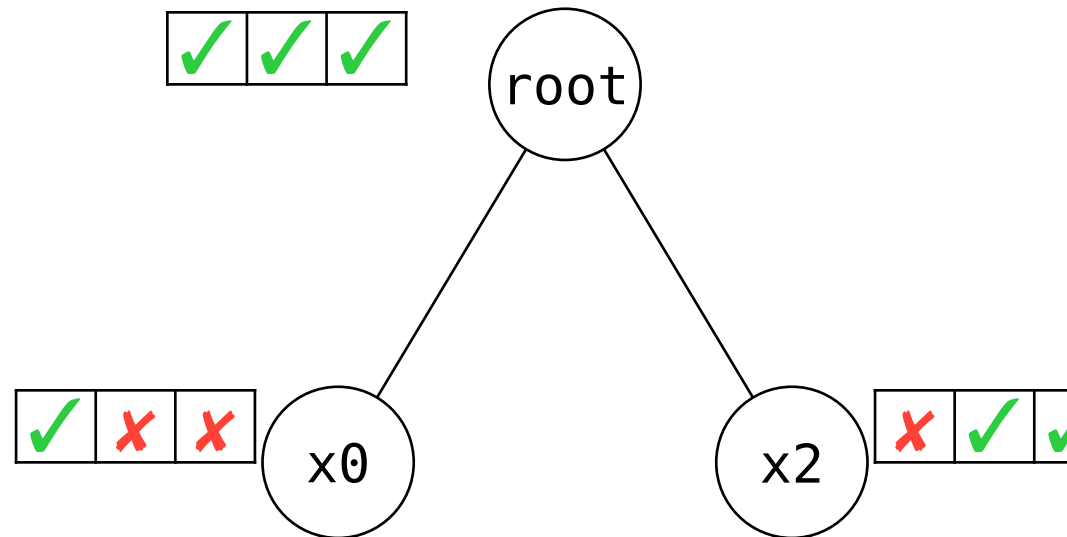
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// Scenario 2

```
unsafe { *x2.sub(1) = 'z'; }
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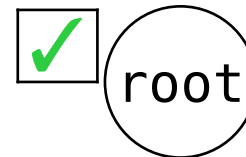
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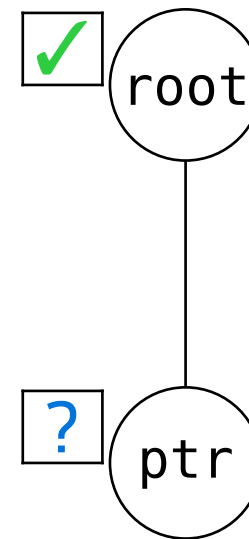


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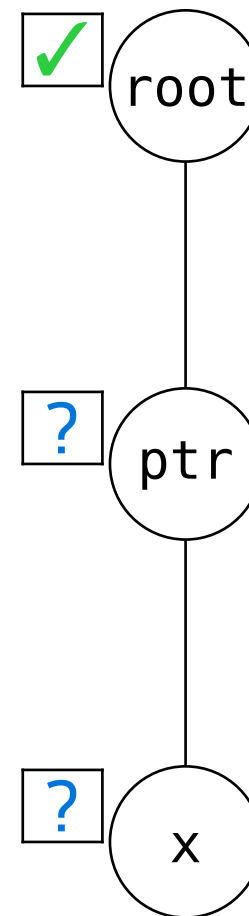
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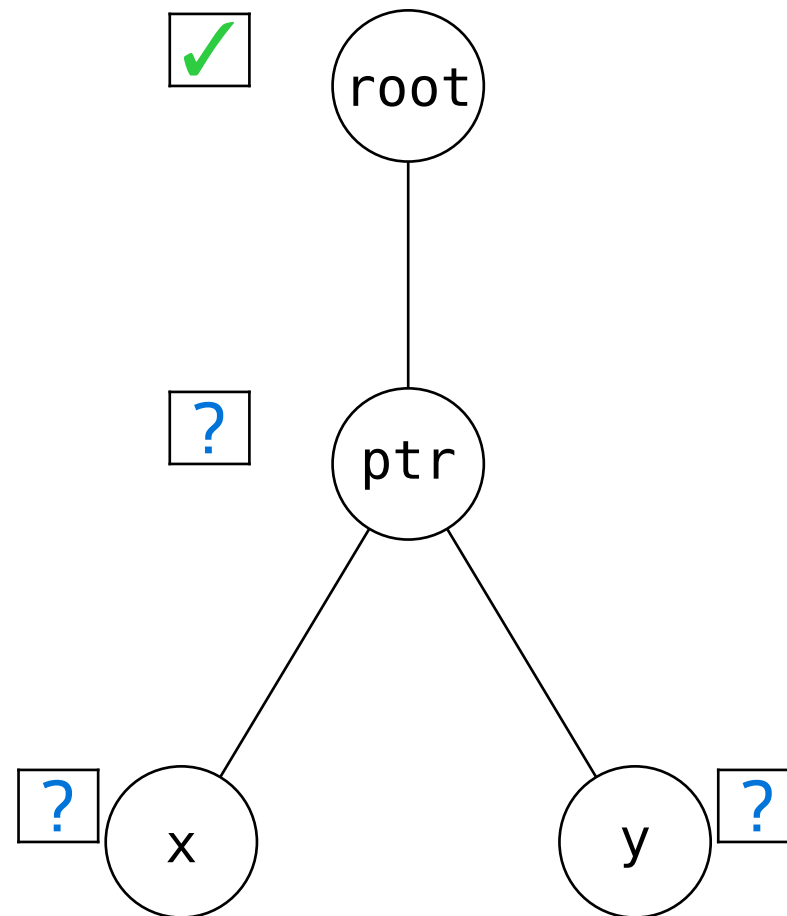


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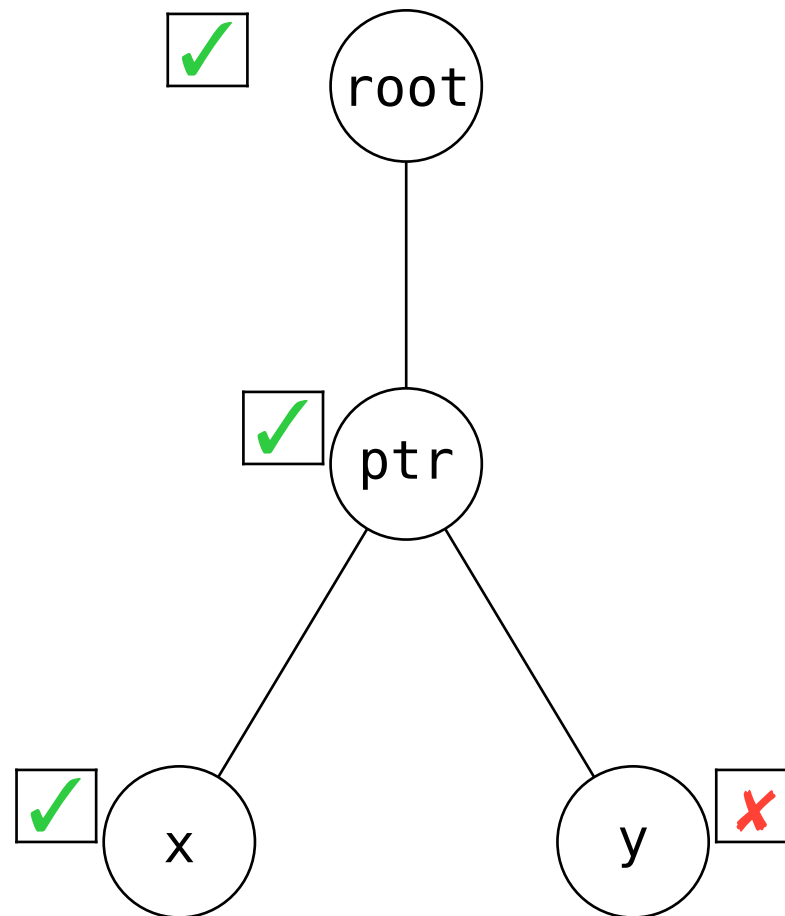


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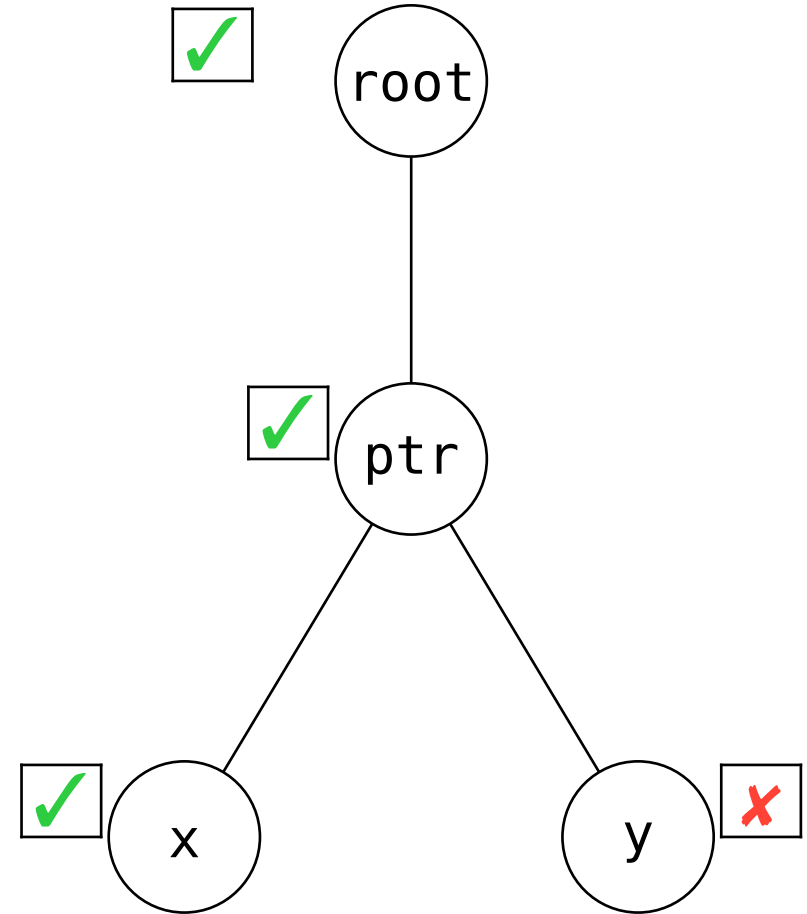
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UB!

Desired outcome: UB



Evaluation

TB should enable desired optimizations

i.e. have enough UB to rule out problematic patterns

Formalized in **Rocq** (+Simuliris), optimizations proven correct

- ✓ delete read through **&mut** or **&**
- ✓ insert read through **&** in function
- ✓ move read down for **&mut** or **&** in function

...

It should be possible to write **unsafe** code free of UB Evaluation

i.e. UB should be predictable and not too common

54% fewer tests have aliasing UB according to Tree Borrows

Only 31 ($< 0.01\%$) tests are regressions, all easily fixable.

(Out of 30 000 libraries, 400 000+ working tests)

“Tree Borrows accepts more real-world programs that call foreign functions than Stacked Borrows due to differences in handling pointer arithmetic.”

A Study of Undefined Behavior Across Foreign Function Boundaries in Rust Libraries,
by I. McCormack, J. Sunshine, J. Aldrich @ ICSE'25

Bonus: Protectors

LLVM attributes get added to the generated code...
is TB sufficient to enforce their requirements ?

- dereferenceable: this pointer is **readable** for the entire function,
- noalias: this pointer has **exclusive access** for the entire function.

Means something should happen on **loss of permissions**.
Not part of the simplified model presented thus far.

```
fn f(x: &mut u64) {  
    *x = 42;  
    opaque()  
    *x  
}
```

Preventing interruptions

```
fn f(x: &mut u64) {  
    opaque()  
    *x = 42  
    *x  
}
```

```
fn f(x: &mut u64) {  
    opaque()  
    *x = 42  
    42  
}
```

Preventing interruptions

```
fn f(x: &mut u64) {  
    *x = 42;  
    opaque()  
    *x  
}
```

```
fn opaque() {  
    let v = unsafe { X };  
    println!("{}", v);  
}
```

```
static mut X: u64 = 0;  
  
fn main() {  
    let raw = &raw mut X;  
    let rmut = unsafe {  
        &mut *raw  
    };  
    f(rmut);  
}
```

Preventing interruptions

```
fn f(x: &mut u64) {  
    opaque()  
    *x = 42  
    *x  
}
```

```
fn opaque() {  
    let v = unsafe { X };  
    println!("{}", v);  
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```

```
static mut X: u64 = 0;  
  
fn main() {  
    let raw = &raw mut X;  
    let rmut = unsafe {  
        &mut *raw  
    };  
    f(rmut);  
}
```

Add a **protector** to `&` and `&mut` function parameters.

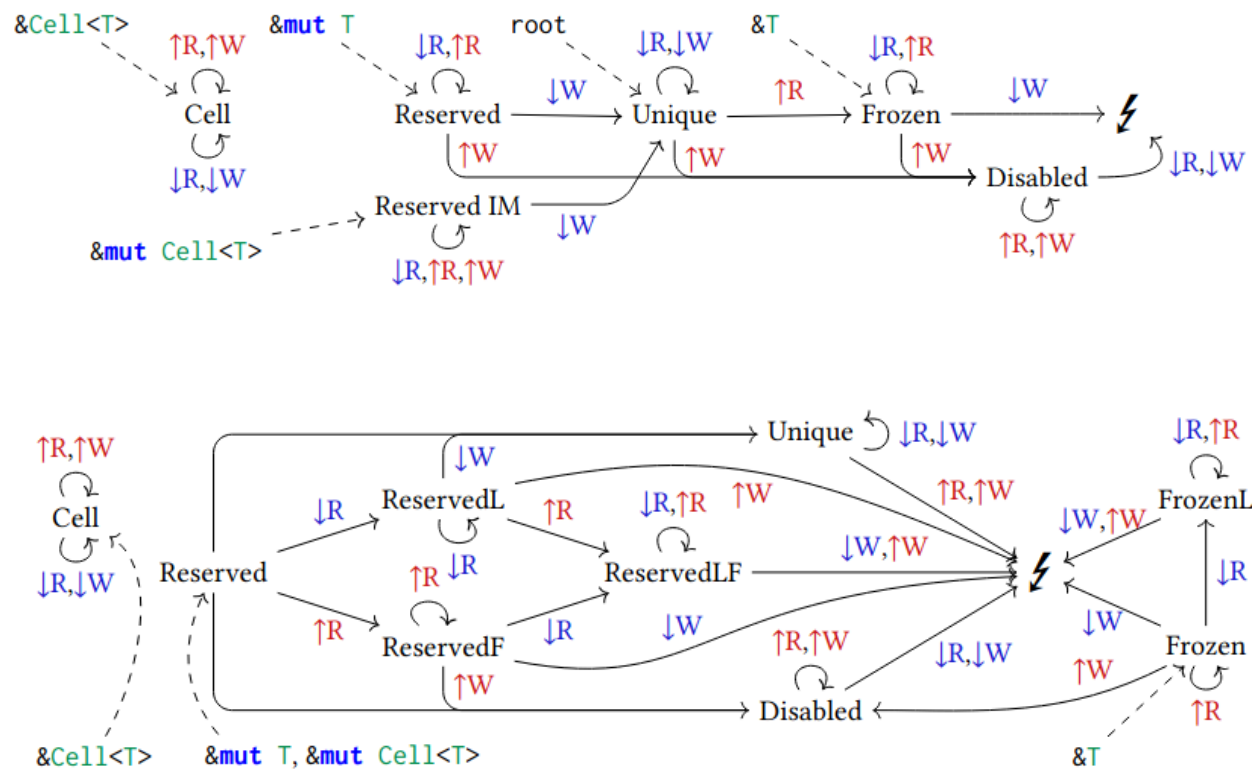
Protector is valid until the function exits, and affects behavior:

- loss of permissions triggers UB,
- more sensitive to foreign accesses.

Other simplifications

I haven't talked about...

- interior mutability
- raw pointers

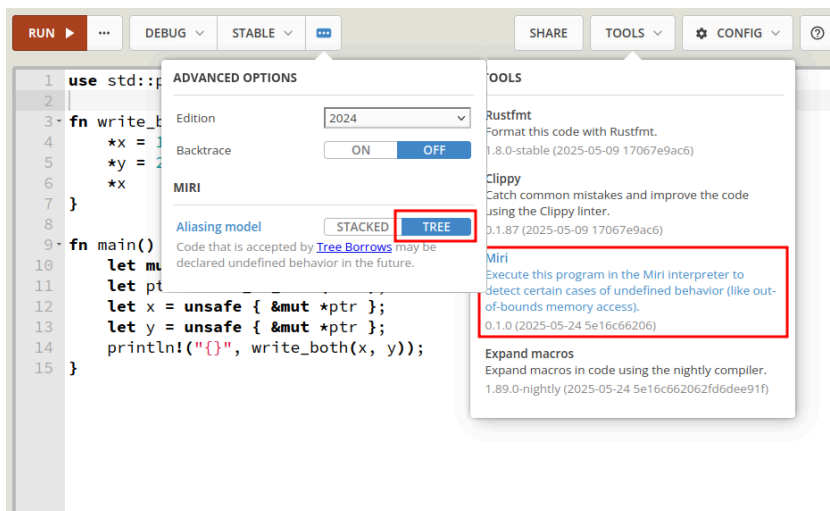


Figures drawn by J. Hostert

Conclusion

Try it out: supported by Miri

Also on the Rust Playground (play.rust-lang.org)



Learn more:

[plf.inf.ethz.ch/research/
pldi25-tree-borrows.html](https://plf.inf.ethz.ch/research/pldi25-tree-borrows.html)



Includes e.g. handling of raw pointers and interior mutability.